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THE YENEPOYA INSTITUTE OF ARTS SCIENCE COMMERCE AND MANAGEMENT

PROJECT-LOW LEVEL DESIGN

On

Visual Threat Hunting for E-commerce Networks

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TABLE OF CONTENT :

1. Introduction ▪▪ **3**

1.1 Purpose and Scope
1.2 Project Executive Summary
1.3 Constraints
1.4 Document Organization

2. System Overview ▪▪**5**

3. Module Design ▪▪**6**

3.1 Data Collection and Loading Module
3.2 Preprocessing Module
3.3 Visualisation Module
3.4 Threat Detection Module
3.5 Reporting and Dashboard Module

4. Data Flow Design ▪▪**7**

5. Database Design ▪▪**7**

5.1 Network Traffic Table
5.2 Derived Detection Table
5.3 Anomaly Alert Table

6. User Interface Design ▪▪**8**

7. Security Implementation ▪▪**8**

8. Testing and Validation ▪▪**8**

9. Project References ▪▪ **9**

10. Glossary ▪▪**9**

11. Front-end ▪▪**10**

12. Back-end ▪▪**13**

1. INTRODUCTION

1.1 Purpose and Scope

Purpose:

This document provides a detailed low-level design of a Visual Threat Hunting system tailored for an E-commerce network environment. The goal is to design a data analytics and visualisation pipeline that detects unusual traffic spikes and identifies potential cyber threats using Python, Pandas, and Seaborn.

The system focuses on demonstrating how network traffic is analysed, how anomalies are detected using rule-based thresholds, and how visualisation helps in cybersecurity threat hunting.

Scope:

The system covers:

- Network traffic data loading and preprocessing using Pandas
- Rule-based anomaly detection using byte threshold algorithm
- Visualisation of traffic patterns using Seaborn and Matplotlib
- Identification of suspicious source IPs and attack patterns
- Interactive web portal for live traffic visualisation and analysis

In Scope:

- Loading and processing e-commerce network traffic CSV dataset
- Rule-based anomaly detection (threshold-based flagging at 1,500 bytes)
- 6 professional visualisation charts (line, bar, heatmap, scatter, dashboard)
- Comprehensive threat hunting dashboard and interactive web portal
- Threat alerts with severity classification (High, Medium, Low)

Out of Scope

- Real-time packet capture from live network interfaces
- Machine learning based anomaly detection
- Payment or transaction security modules
- Integration with external SIEM or firewall tools

1.2 Project Executive Summary

Modern E-commerce platforms handle thousands of network transactions per second, making them prime targets for cyber attacks such as Distributed Denial of Service (DDoS) and data exfiltration. Identifying these threats manually without automated visual tools is impractical.

This project introduces a structured Visual Threat Hunting system that:

- Analyses e-commerce network traffic data using Pandas DataFrames
- Applies rule-based threshold detection to flag anomalies above 1,500 bytes
- Produces 6 distinct visualisations using Seaborn and Matplotlib
- Generates an automated Threat Hunting Summary Report in Jupyter Notebook
- Provides an interactive web portal for live network analysis and visualisation

The system is implemented using:

- **Language:** Python 3.9+, Pandas, Seaborn, Matplotlib
- **Dashboard:** Jupyter Notebook (Python Environment)
- **Storage:** CSV File (network_traffic.csv)
- **Frontend:** HTML, CSS, JavaScript, Chart.js

This solution demonstrates how visual threat hunting works in an E-commerce context and provides a working network traffic analysis and threat intelligence dashboard.

1.3 Constraints

Technical Constraints

- Dataset is simulated; real-world deployment requires live packet capture tools
- Rule-based threshold detection may produce false positives for unusual legitimate traffic
- Seaborn visualisations are static; real-time updates require additional tooling
- Jupyter Notebook with Python 3.9+ required to run the analysis pipeline

Security Constraints

- Dataset contains simulated IP addresses for academic demonstration purposes
- Threshold value must be carefully calibrated to avoid missing genuine threats
- No user authentication implemented in the current portal version

Operational Constraints

- Designed for academic and demonstration purposes
- Limited to single-day traffic analysis in current dataset
- Requires stable Python environment with Pandas and Seaborn libraries installed

1.4 Document Organization

The document is structured into:

1. Introduction
2. System Overview
3. Module Design
4. Data Flow
5. Database Structure
6. Interface Design
7. Security Implementation
8. Testing Strategy

2. System Overview

The system is designed using a data analytics pipeline architecture:

- Users interact through Jupyter Notebook interface and web browser portal
- Python processes, analyses, and visualises the e-commerce network traffic data
- CSV dataset stores all 48 network traffic records for offline analysis
- Interactive portal reads the data and presents it visually in real time

The system ensures that each traffic record is evaluated against the defined 1,500 byte threshold before being classified as normal or anomalous, making threat patterns completely visible to the security analyst.

3. Module Design

3.1. Data Collection and Loading Module

- Loads e-commerce network traffic CSV dataset using Pandas read_csv()
- Parses Timestamp column automatically for time-based analysis
- Validates dataset structure and checks for missing or malformed values
- Displays dataset shape, column names, and descriptive statistics

3.2. Preprocessing Module

- Extracts Hour and Time_Label features from the Timestamp column
- Creates binary Is_Anomaly flag derived from the Label column
- Groups data by hour and protocol for aggregated time-based analysis
- Applies rule-based detection threshold to classify each traffic record

3.3. Visualisation Module

- Plot 1 — Traffic Timeline: Line chart of bytes over 48 windows with anomalies in red
- Plot 2 — Threat Detection: Colour-coded bar chart with normal vs flagged count
- Plot 3 — Protocol Analysis: Bar chart and box plot of bytes per protocol
- Plot 4 — Heatmap: Seaborn heatmap of average bytes by protocol and hour
- Plot 5 — IP Analysis: Source IP ranking by bytes with anomaly scatter plot
- Plot 6 — Dashboard: 6-panel composite dashboard combining all visualisations

3.4. Threat Detection Module

- Implements rule: IF Bytes_Sent > THRESHOLD (1500) THEN Label = Threat Detected
- Default threshold 1,500 bytes set based on baseline traffic analysis (~820 bytes avg)
- Identifies 3 attack types: DDoS Attack, Data Exfiltration, Isolated Traffic Spike
- Generates flagged records list with source IP, timestamp, protocol, and peak bytes

3.5. Reporting and Dashboard Module

- Prints detailed Threat Hunting Summary Report to Jupyter Notebook output
- Saves all 6 visualisation plots as high-resolution PNG files
- Provides interactive web portal with 4 analysis tabs via HTML/JavaScript
- Displays live metric cards, adjustable threshold slider, and threat alert panels

4. Data Flow Design

i. Userpl aces network_traffic.csv and notebook in the same project folder
ii. Jupyter Notebook loads the CSV dataset using Pandas read_csv()
iii. Preprocessing extracts time features and creates anomaly flag column
iv. Rule-based detection algorithm applies 1,500 byte threshold to each record
v. Each record is classified as Normal or Threat Detected
vi. Seaborn and Matplotlib generate 6 visualisation plots
vii. All plots are saved as PNG files in the project folder
viii.Threat Hunting Summary Report is printed to notebook output

5. Database Design

Storage Used: CSVFile(network_traffic.csv)

Tables Used

Network Traffic Table

- Timestamp
- Bytes_Sent
- Bytes_Received
- Packets
- Protocol
- Source_IP
- Destination_IP
- Port
- Duration_ms
- Label

Derived Detection Table

- Hour
- Time_Label
- Is_Anomaly
- Detected_Threat

Anomaly Alert Table

- Source_IP
- Peak_Bytes
- Attack_Type
- Severity
- Timestamp

6. User Interface Design

The system includes:

- Jupyter Notebook interface — step-by-step code cells with inline visualisations
- Traffic Timeline Chart with adjustable anomaly detection threshold
- Threat Detection Dashboard (6-panel composite view)
- Interactive Web Portal (threat_hunting_portal.html)
- Threat Alerts Panel with severity classification (High, Medium, Low)

The interface is designed with a clean data-analytics theme for professional E-commerce cybersecurity threat monitoring.

7. Security Implementation

- Rule-basedanomaly detection flags all records exceeding the 1,500 byte threshold
- Source IP analysis identifies external suspicious IPs vs internal network IPs
- Severity classification (High, Medium, Low) applied to each detected threat
- Timestamps recorded for all anomaly events to support forensic timeline analysis
- All flagged IP addresses isolated in a dedicated threat summary table

8. Testing and Validation

- Functional testing of data ~~loading and preprocessing pipeline —~~
- Threshold detection validation — 6 anomalies correctly flagged out of 48 records (12.5%)
- Visualisation output testing — all 6 plots generated and saved successfully as PNG
- Portal tab navigation testing — all 4 tabs render correctly in browser
- Basic security testing — flagged IPs validated against dataset ground truth labels

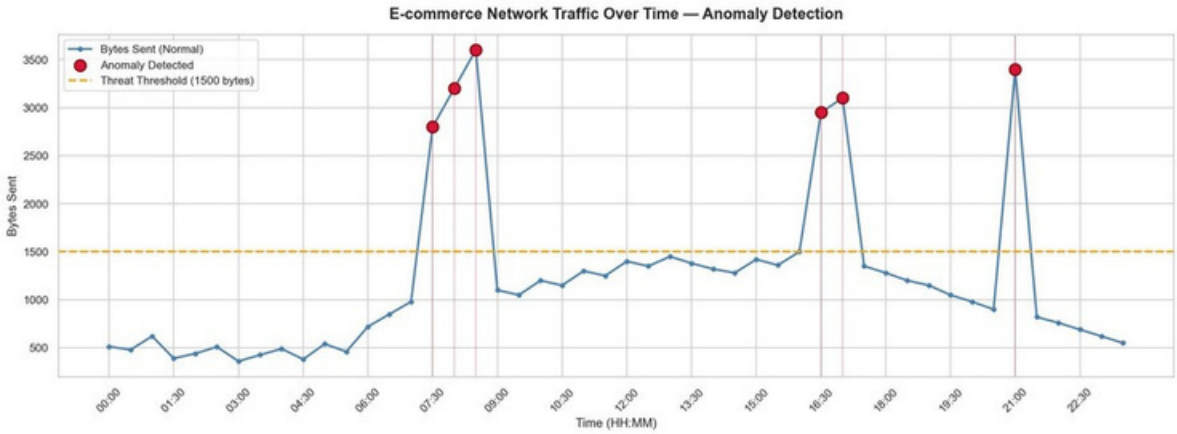
9. Project References

• Python Software Foundation — Python 3.9 Official Documentation
• Pandas Development Team — Pandas Documentation. https://pandas.pydata.org/docs/
• Waskom, M. (2021) — Seaborn: Statistical Data Visualization. Journal of Open Source Software
• OWASP Foundation — OWASP Top 10 Security Guidelines. https://owasp.org/Top10/
• NIST — Guide to Intrusion Detection and Prevention Systems. SP 800-94
• Jupyter Notebook — https://jupyter-notebook.readthedocs.io/

10. Glossary

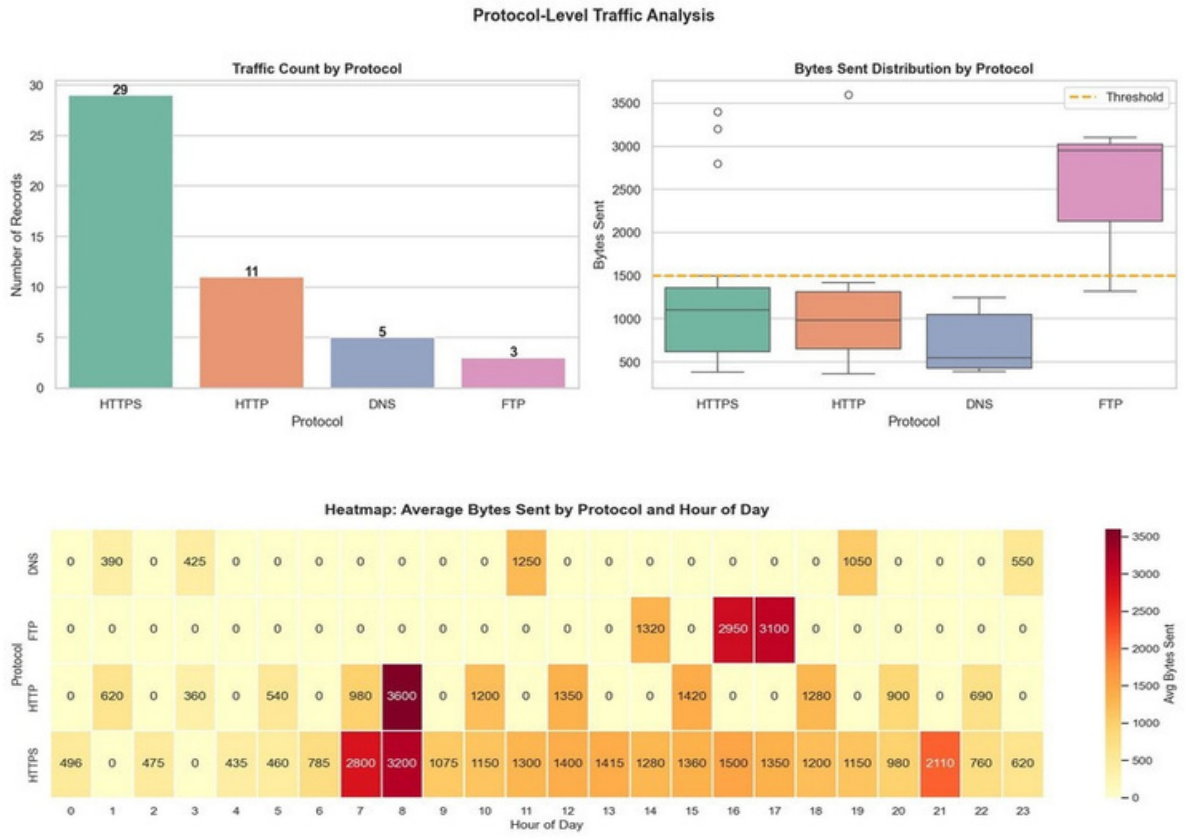
• **DDoS:** Distributed Denial of Service — floods a server with excessive traffic to cause unavailability
• **Threat Hunting:** Proactive search through network data to detect hidden cyber threats
• **Threshold:** A defined byte value above which traffic is classified as anomalous
• **Anomaly:** A traffic record deviating significantly from the established baseline pattern
• **Pandas:** Python library for data manipulation and analysis using DataFrames
• **Seaborn:** Python statistical data visualisation library built on Matplotlib
• **Heatmap:** Colour intensity matrix used to represent data value distributions
• **Protocol:** Set of rules governing network communication (HTTPS, HTTP, FTP, DNS)
• **Source IP:** Originating IP address of a network connection
• **Baseline:** Normal expected traffic level used as reference for anomaly detection
• **Threat Intel:** Information collected about cyber attack patterns and behaviours

FRONT-END

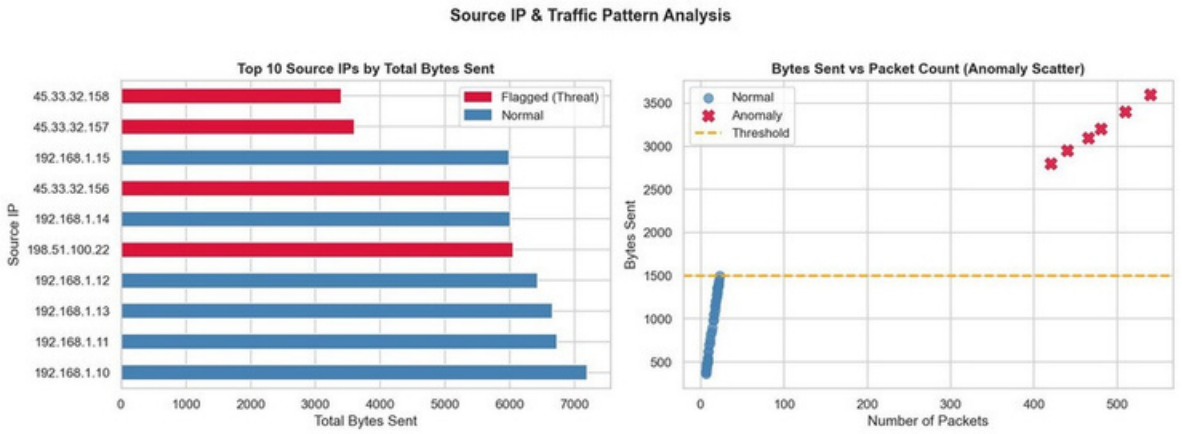

E-commerce Network Traffic Over Time — Anomaly Detection
Bytes Sent (Normal)
Anomaly Detected
Threat Threshold (1500 bytes)
Bytes Sent
Time (HH:MM)

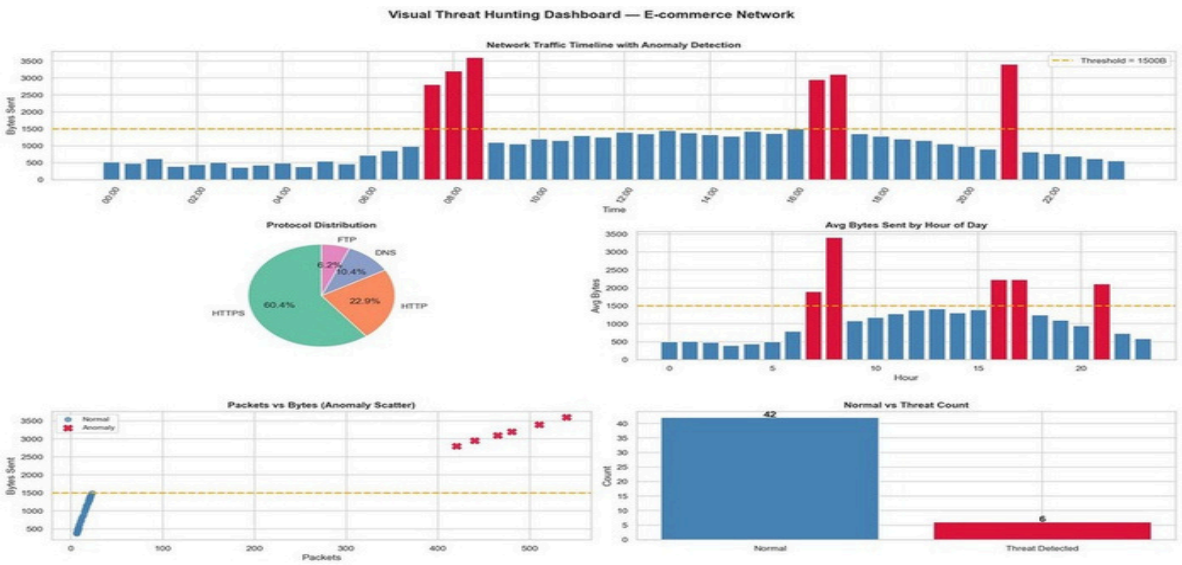
Rule-Based Threat Detection Results
Traffic Bytes — Threats Highlighted in Red
Threshold (1500B)
Bytes Sent
Time
Normal vs Threat Count
42
6
Number of Records
Normal
Threat Detected
Traffic Status

FRONT-END (continued)



FRONT-END (continued)





BACK-END – PYTHON

Technology Stack:

- Language : Python 3.9+
- Data Analysis : Pandas 1.5+
- Visualisation : Seaborn 0.12+, Matplotlib 3.6+
- Environment : Jupyter Notebook
- Portal Frontend : HTML5, CSS3, JavaScript, Chart.js 4.4.1
- Dataset : network_traffic.csv (48 records, 10 columns)

Jupyter Notebook Code Structure:

Step 1 — Import Libraries

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

Step 2 — Load Dataset

```
df = pd.read_csv('network_traffic.csv', parse_dates=['Timestamp'])
```

Step 3 — Preprocessing

```
df['Hour'] = df['Timestamp'].dt.hour
df['Is_Anomaly'] = df['Label'].apply(lambda x: 1 if x=='Anomaly' else 0)
```

Step 4 — Anomaly Detection Rule

```
THRESHOLD = 1500
df['Detected_Threat'] = df['Bytes_Sent'].apply(
    lambda x: 'Threat Detected' if x > THRESHOLD else 'Normal')
```

Step 5 — Plot 1: Traffic Timeline

Step 6 — Plot 2: Threat Detection Bar Chart

Step 7 — Plot 3: Protocol Analysis

Step 8 — Plot 4: Seaborn Heatmap

```
sns.heatmap(heatmap_data, annot=True, fmt='.0f', cmap='YlOrRd')
```

Step 9 — Plot 5: Source IP & Traffic Pattern Analysis

Step 10 — Threat Hunting Summary Report

Step 11 — Plot 6: 6-Panel Dashboard

Step 12 — Interactive Web Portal (HTML embedded)